Initial Post

by Anne Dayer - Saturday, 25 October 2025, 11:04 PM

Number of replies: 5

The Evolution of Enterprise IT in Healthcare as Societal Infrastructure

Since the 1990s, enterprise computing in healthcare has evolved from isolated administrative systems to national-scale digital infrastructure. This transformation reflects the broader shift in computing from automation of internal processes to strategic enablers of public service delivery, clinical coordination, and societal resilience.

The Enterprise IT Body of Knowledge (EITBOK) defines enterprise IT as dependent on aligned governance, shared decision rights, and platform scalability (EITBOK, no date). These principles are increasingly evident in the National Health Service (NHS) in England, where cloud-based systems now support real-time access to patient records, digital diagnostics, and service integration across organisations (NHS England, 2023).

Two major benefits are visible. First, real-time data availability supports more consistent clinical decisions, particularly in emergency settings. Second, cloud infrastructure reduces reliance on fragmented local systems, allowing resources to be redirected from IT maintenance toward frontline care. NHS Digital notes that the GP IT Futures framework expired in March 2023, and practices are now transitioning to new cloud-capable systems under the Digital Services for Integrated Care model (NHS Digital, 2023).

However, growing reliance on centralised digital systems introduces significant risks. A 2023 ransomware attack delayed cancer diagnostics and disrupted emergency care, demonstrating how digital failure directly threatens patient safety (Department of Health and Social Care & NHS England, 2023). At the same time, over two million UK residents experience digital exclusion—particularly older people and rural communities—raising concerns that digital-first health strategies may unintentionally widen inequality (Good Things Foundation, 2022).

Walton (2024) highlights that computing progress is not neutral; its benefits are unequally distributed. This underscores a critical governance tension: NHS dependence on centralised IT procurement may contradict EITBOK's ideal of shared decision rights and local adaptability, risking fragile systems and diminished accountability.

Reflecting on this, I now recognise enterprise IT as more than a technical backbone. It is a computing infrastructure that defines access, equity, and trust in public health. As such, its design and governance must balance innovation with inclusion to deliver true societal value.

References

Department of Health and Social Care & NHS England (2023) *A cyber-resilient health and adult social care system in England: Cyber security strategy to 2030*. Available at: **https://www.gov.uk/government/publications/cyber-security-strategy-for-health-and-social-care-2023-to-2030/a-cyber-resilient-health-and-adult-social-care-system-in-england-cyber-security-strategy-to-2030** (Accessed: 27 October 2025).
Enterprise IT Body of Knowledge (EITBOK) (no date) *Enterprise IT Body of Knowledge*. Available at: **https://eitbokwiki.org** (Accessed: 27 October 2025).
Good Things Foundation (2024) *Digital inclusion and health statistics*. Available at: **https://www.goodthingsfoundation.org/discover/digital-inclusion-insights/digital-inclusion-**

and-health (Accessed: 27 October 2025).
NHS Digital (2023) *GP IT Futures systems and services*. Available at: **https://digital.nhs.uk/services/gp-it-futures-systems** (Accessed: 27 October 2025).
NHS England (2023) *Digital Transformation Programme Overview*. Available at: **https://www.england.nhs.uk/digitaltechnology** (Accessed: 27 October 2025).
Walton, D.J. (2024) *Culturally Responsive Computing: An Introduction into Computer Science, Security, and Technology*. Boston, MA: ROTEL Publications.

In reply to Anne Dayer

Re: Initial Post

by Tugba Unal - Tuesday, 28 October 2025, 2:30 PM

Thank you for your insightful discussion on the evolution of enterprise IT within healthcare. You clearly illustrate how computing has progressed from administrative support to becoming a crucial enabler of public service delivery and clinical integration. The connection you make between the Enterprise IT Body of Knowledge (EITBOK) principles—governance alignment, shared decision rights, and scalability—and the NHS's ongoing digital transformation is particularly well observed.

Your analysis of the benefits of real-time data access and cloud-based infrastructure effectively demonstrates how enterprise IT can enhance clinical decision-making and operational efficiency. However, I appreciate your balanced perspective on the associated risks, particularly cybersecurity threats and digital exclusion. The example of the 2023 ransomware attack powerfully highlights the vulnerabilities of centralised systems, while your reference to digital inequality emphasises the ethical responsibility to ensure equitable access for all.

I also value your linkage to Walton's (2024) perspective, which reinforces that computing outcomes are shaped by social contexts and power dynamics. Preserving shared decision rights and local adaptability will be vital to avoid governance fragility and sustain public trust.

Ultimately, your conclusion rightly underscores that enterprise IT is not merely technical infrastructure—it is a socio-technical system defining access, equity, and confidence in digital healthcare.

References:
Walton, D.J. (2024) Culturally Responsive Computing: An Introduction into Computer Science, Security, and Technology. Boston, MA: ROTEL Publications.

In reply to Tugba Unal

Re: Initial Post

by Anne Dayer - Saturday, 1 November 2025, 4:25 PM

Thank you, Tugba, for your thoughtful engagement with my post. I appreciate how you summarised the governance and inclusion aspects and your closing remark on enterprise IT as a socio-technical system defining access, equity, and confidence in healthcare. I'll reflect that terminology in my summary work.

In reply to Anne Dayer

Re: Initial Post

by [Thomas Shadforth]() - Monday, 3 November 2025, 10:01 PM

Peer Response

Hello Anne, it was great to read your piece discussing the gradual evolution of Enterprise IT within the Healthcare field. It is important to consider what impact this particular development has had within recent years and how; moving forward, it has the potential to be shaped in a number of different ways based on numerous external factors.

In reference to one of these factors, it is very important to highlight Walton's (2024) discussion on how different aspects of computing not only has an impact on the cultures in which it is employed, but how cultural influences can impact the overall design of a technological solution, ranging from the cultural approaches to programming and problem solving, to how inclusive and accessible computing technology can be for a given audience. Walton states that "An awareness of how cultural factors influence computing helps us build more inclusive, effective and sensitive technological solutions", highlighting the importance of how societal culture can play a big part in shaping computing.

It is wise to factor in these cultural influences and their everchanging nature, as future implementations can create a vastly different solution from those that are in use today.

Reference

Walton, D.J. (2024) *Culturally Responsive Computing: An Introduction into Computer Science, Security and Technology.* Boston, MA: ROTEL Publications

---

In reply to Thomas Shadforth

Re: Initial Post

by [Anne Dayer]() - Wednesday, 5 November 2025, 9:12 AM

Thank you, Thomas, for highlighting Walton's point on cultural influences. Your observation that societal culture shapes both design and inclusivity adds an important dimension I hadn't fully addressed. I'll integrate this perspective into my summary—appreciate the insight.

In reply to Anne Dayer

Summary Post

by [Anne Dayer]() - Monday, 10 November 2025, 4:13 PM

My understanding of enterprise IT in healthcare has shifted considerably through this discussion. Initially, I saw enterprise IT as technical infrastructure; I now understand it as a contested socio-technical system shaped by governance choices, cultural contexts, and structural inequalities. My original post examined NHS digital transformation through EITBOK's governance lens, weighing clinical coordination benefits against cybersecurity vulnerabilities and digital exclusion. Tugba reframed enterprise IT as defining "access, equity, and confidence" rather than merely efficiency, introducing socio-technical systems thinking that had not featured in my analysis. Thomas connected this to Walton's (2024) work on culturally responsive computing, which argues that design choices reflect the values and power structures of those creating them. These contributions challenged my assumption that computing solutions could remain neutral. The module content deepened this rethinking. I have come to recognise that technical architecture and governance architecture are fundamentally intertwined. Unit 2 shows how Boolean logic gates establish which computational pathways become possible and which do not. Scaled to enterprise level, these encode exclusionary logics determining system access and capabilities. Unit 4's software engineering principles reveal a structural tension: the NHS's move toward vendor-led cloud platforms potentially contradicts EITBOK's emphasis on shared decision rights, creating dependency structures that limit local adaptation (EITBOK, no date; Stephens, 2022). NHS ransomware attacks have demonstrated how centralised architectures concentrate vulnerabilities (Department of Health and Social Care & NHS England, 2023). NHS England defends centralisation through standardisation and economies of scale (NHS Digital, 2023). Yet healthcare as a public good requires democratic accountability, not merely administrative efficiency. Digitally excluded groups, disproportionately elderly, disabled, and low-income populations, face substantial barriers accessing NHS services, with 33% of offline users finding it difficult to interact with NHS digital systems (Good Things Foundation, 2024). My perspective has moved from technical confidence toward critical humility, recognising enterprise IT as a contested space where questions about who designs, who decides, and who benefits cannot be separated from technical choices.

References

Department of Health and Social Care and NHS England (2023) *A cyber-resilient health and adult social care system in England: Cyber security strategy to 2030*. Available at: **https://www.gov.uk/government/publications/cyber-security-strategy-for-health-and-social-care-2023-to-2030/a-cyber-resilient-health-and-adult-social-care-system-in-england-cyber-security-strategy-to-2030** (Accessed: 10 November 2025).

Enterprise IT Body of Knowledge (EITBOK) (no date) *Enterprise IT Body of Knowledge*. Available at: **https://eitbokwiki.org** (Accessed: 10 November 2025).

Good Things Foundation (2024) *Digital inclusion and health statistics*. Available at: **https://www.goodthingsfoundation.org/discover/digital-inclusion-insights/digital-inclusion-insights-2024/digital-inclusion-and-health** (Accessed: 10 November 2025).

NHS Digital (2023) *GP IT Futures systems and services*. Available at: **https://digital.nhs.uk/services/gp-it-futures-systems** (Accessed: 10 November 2025).

Stephens, R. (2022) *Beginning Software Engineering*. 2nd edn. Hoboken, NJ: Wiley.

Walton, D.J. (2024) *Culturally Responsive Computing: An Introduction into Computer Science, Security, and Technology*. Boston, MA: ROTEL Publications.